

SDLC Models

1. > waterfall Model : it is the simplest Model of s/w development paradigm. it says that all the phases of SDLC will

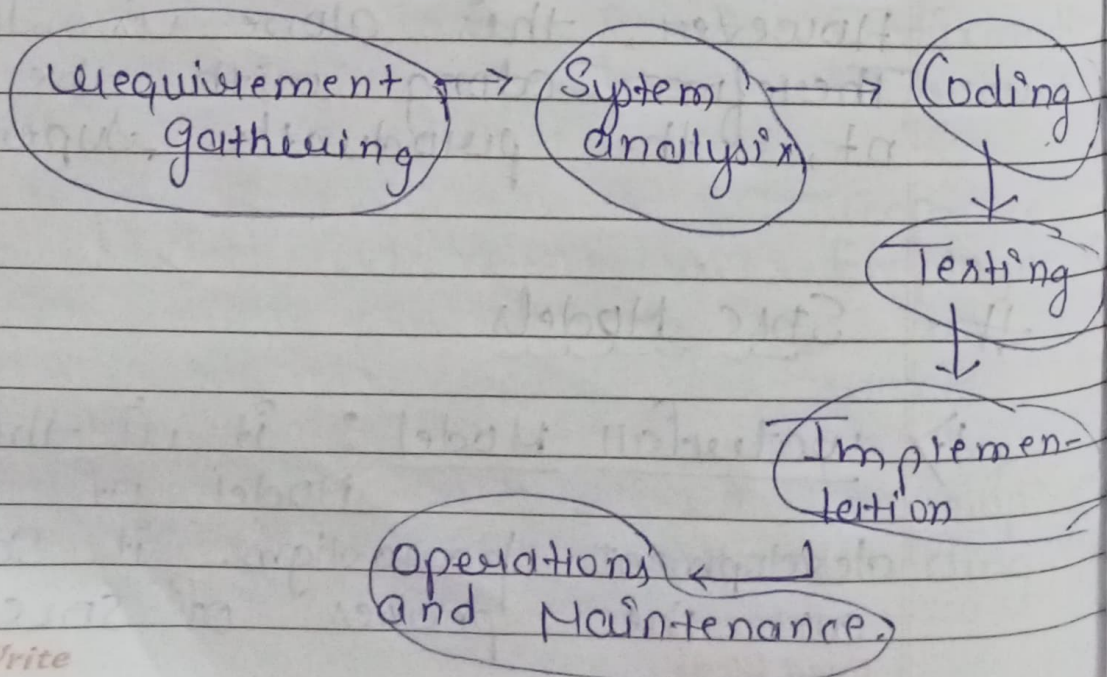
Good Write

Function one after another in linear manner.

That is, when the 1st phase is finished then only the 2nd phase will start and so on.

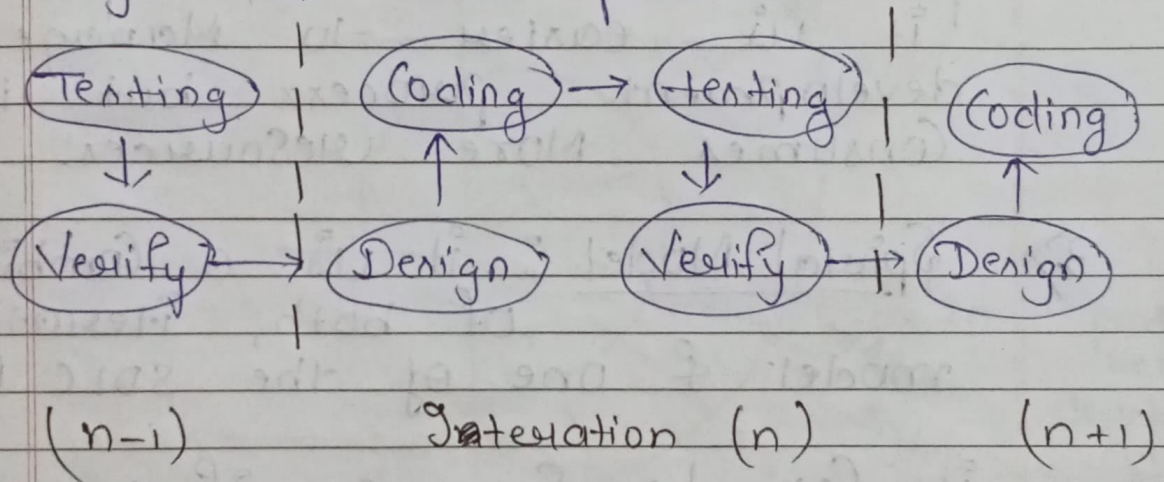
This Model assumes that everything is carried out & taken place perfectly as planned in the previous stage and there is no need to think about the past issues that may arise in the next phase. This Model does not work smoothly if there are some issues left at the previous step.

The Sequential nature of Model does not allow us to go back & undo or redo our actions.



This Model is best Suited when developers already have designed & developed similar sw in the past & are aware of all its domain.

2. > Iterative - waterfall Model: This Model leads the sw development process in iterations. It projects the process of development in cyclic manner repeating every step after every cycle of SDLC process.



The sw is first developed on Very Small Scale & all the steps are followed which are taken into Consideration. Then, on every next iteration, More features and Modules are designed, Coded, -tested & added to the sw.

every cycle produce a sw, which is complete in itself and has more features and capabilities than that of the previous one.

after each iteration, the Management team can do work on risk Management & prepare for the next iteration.

because a cycle includes small portion of whole sw process, it is easier to manage the development process but it consumes more resources.

> Spiral Model : it is a combination of both, iterative model & one of the SDLC Model.

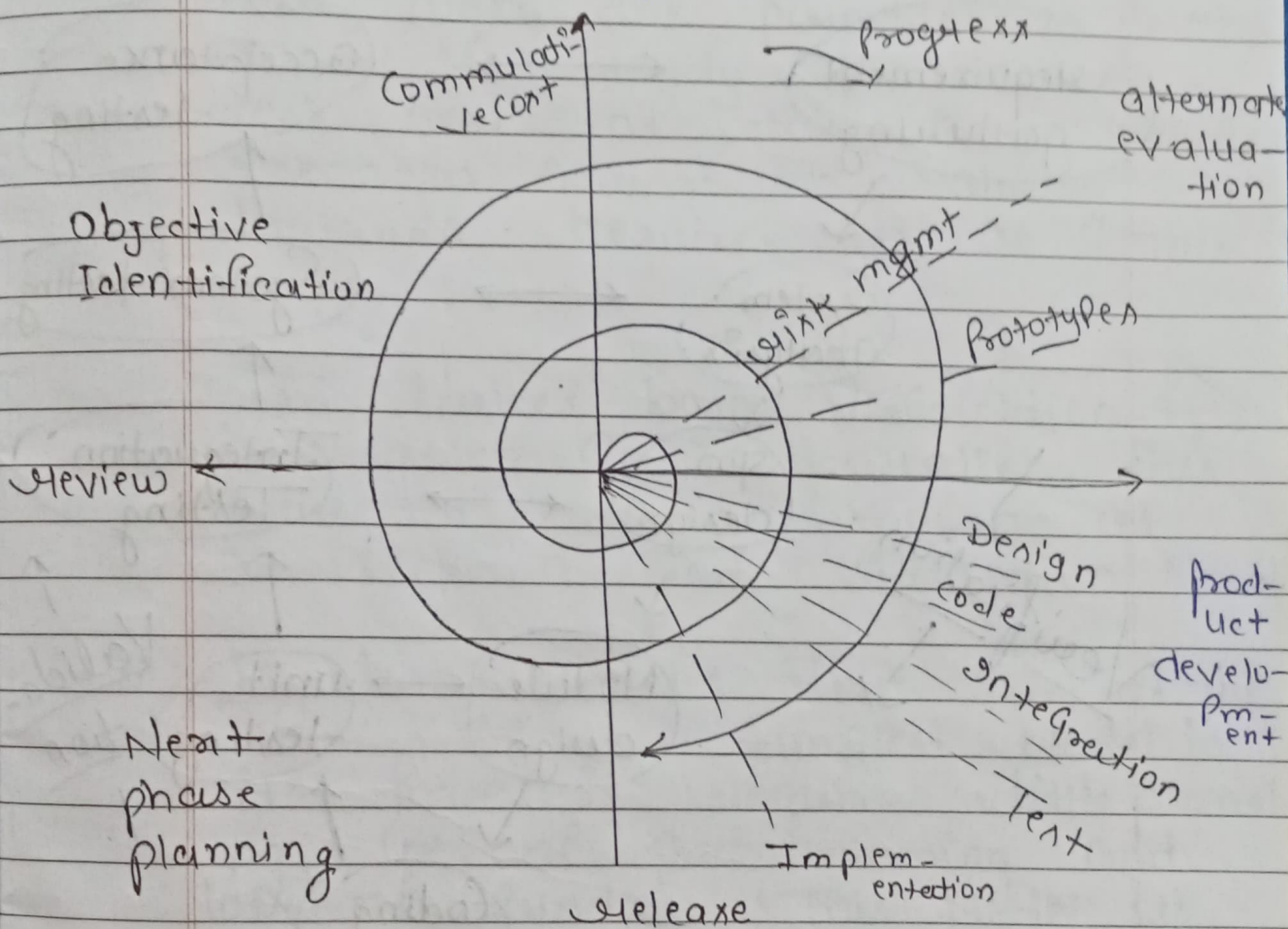
it can be seen as if you choose one SDLC Model and combine it with cyclic process.

This Model considers risk, which often goes unnoticed by most other Models.

The Model starts with determining objectives & constraints

of SW at the start of one iteration. Next phase is of prototyping the SW. This includes risk analysis.

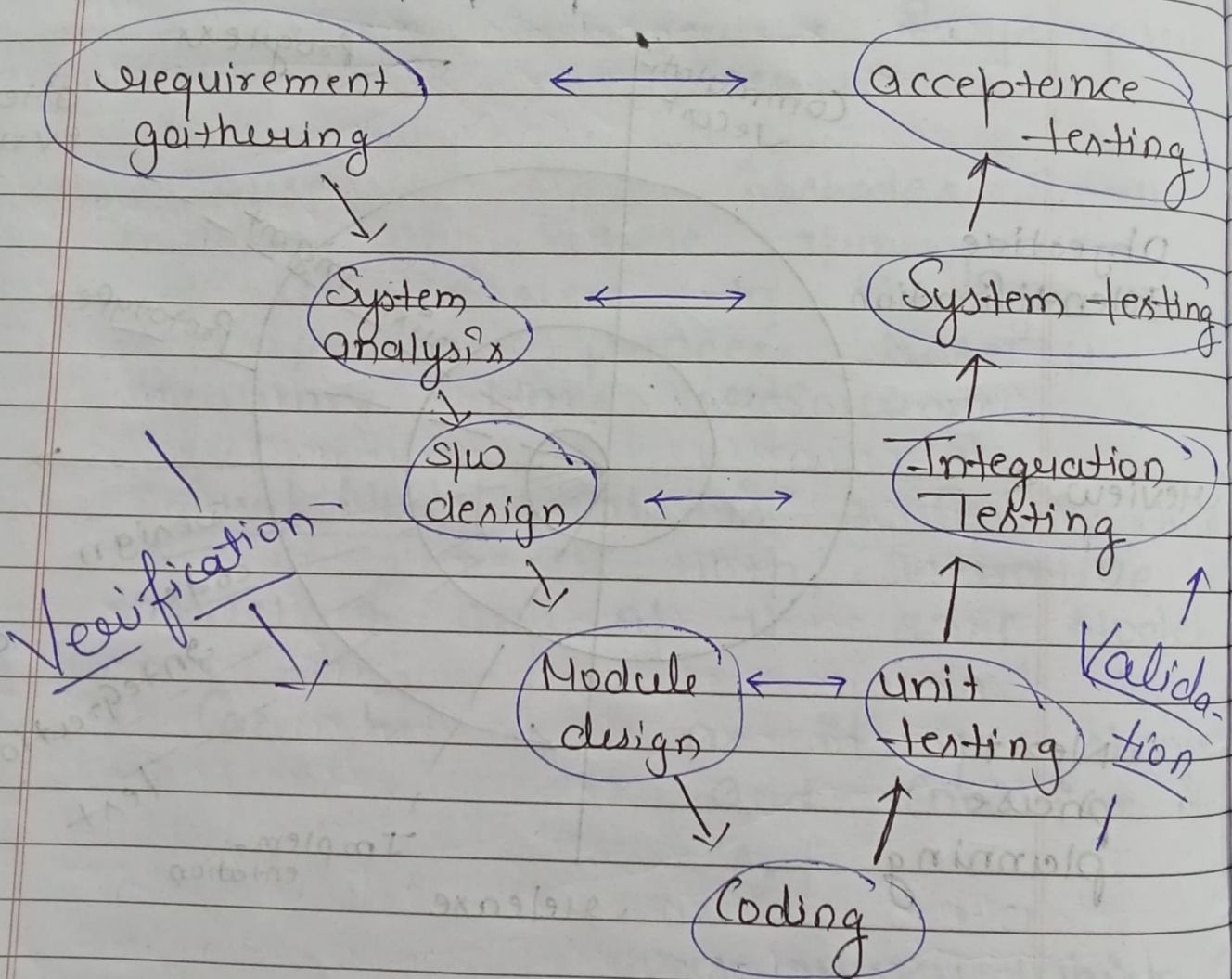
Then one standard SDLC Model is used to build the SW. In the 4th phase of plan of next iteration is prepared.



4. > V-model: The Major drawback of waterfall model is we move to the next stage only when the previous one is finished and there was no

chance to go back if something is found wrong in later stages.

V-Model provides Means of Testing of SW at each stage in reverse manner.



At every stage, test plans & test cases are created to verify and validate the product

DATE: / /
PAGE:
according to the requirement of that stage.

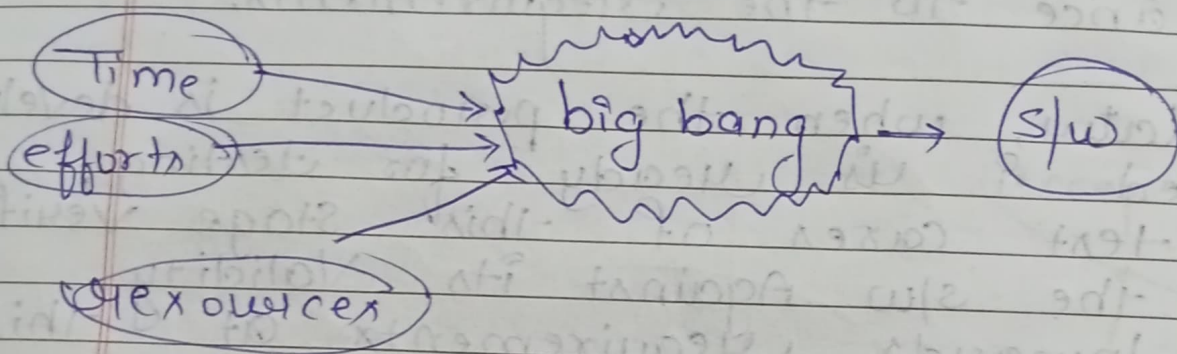
for example, in requirement gathering stage the test team prepares all the test cases in correspondence to the requirements.

later, when the product is developed & is ready for testing, test cases of this stage verify the sw against its validity towards requirements at this stage.

This makes both Verification & Validation go in parallel. This Model is also known as Verification and Validation Model.

5. > big bang Model: This Model is the simplest Model in its form. it requires little planning, lots of programming and lots of funds. This Model is conceptualized around the big bang of universe. As Scientists say that after big bang lots of galaxies, planets and stars evolved just as an event.

likewise, if we put together lots of programming & funds, you may achieve the best s/w product.



for this Model, very small amount of planning is required, it does not follow any process or at times the customer is not sure about the requirements and future needs. So the input requirements are constant.

This Model is not suitable for large s/w projects but good one for learning & experimenting.

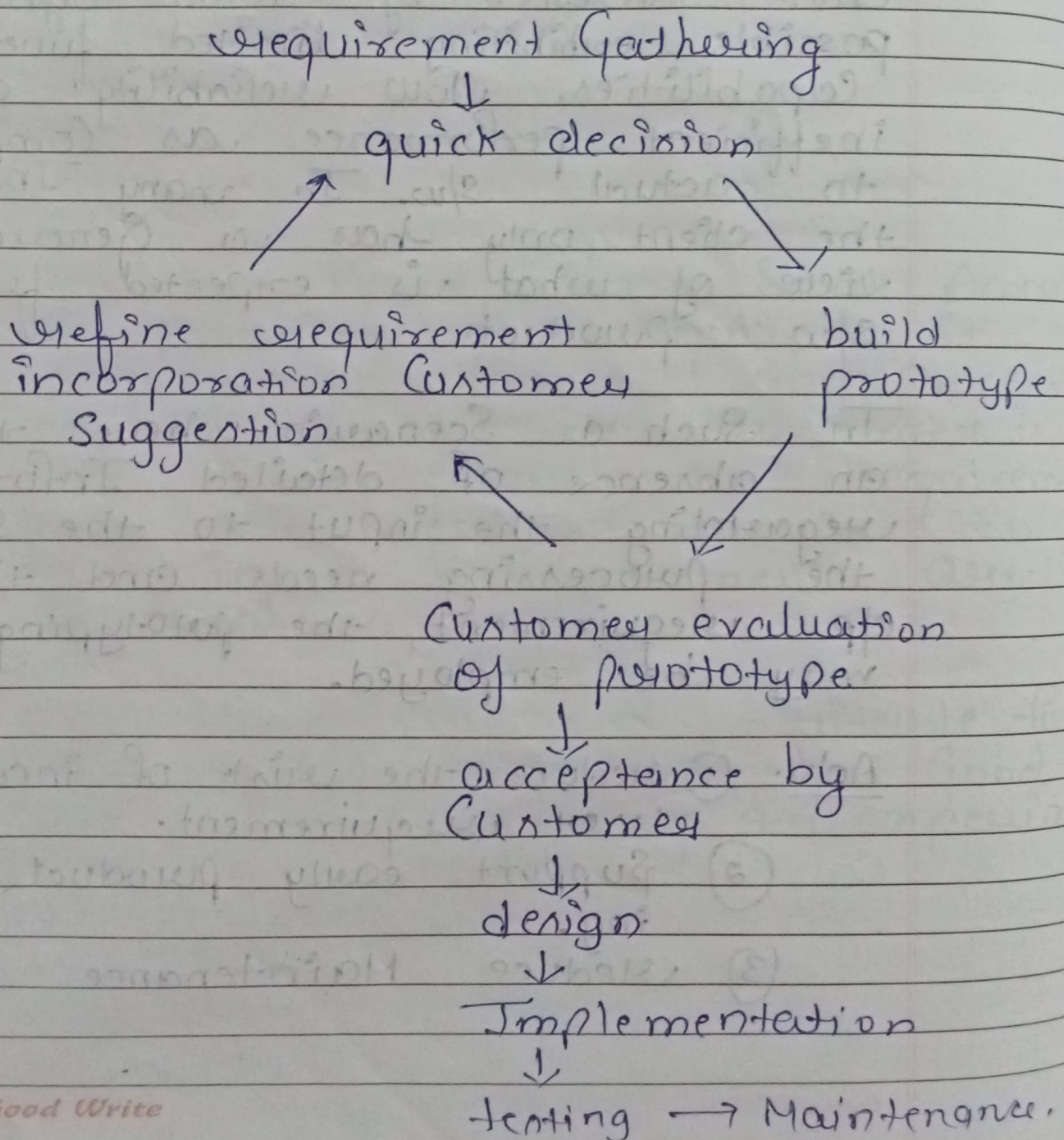
6.) Prototype Model: it requires ~~no~~ that before carrying out the development of actual S/w, a working prototype of the system should be built.

A prototype is a toy implementation of the system. A prototype usually turns out to be a very crude version of actual system, possibly exhibiting limited functional capabilities, low reliability and inefficient performance as compared to actual S/w. In many instances, the client only has a general view of what is expected from the S/w product.

In such a scenario where there is an absence of detailed information regarding the input to the system, the processing needs and the output requirement, the prototyping model may be employed.

- Adv.
- ① Reduce the risk of incorrect user requirement.
 - ② Support early product Marketing.
 - ③ Reduce Maintenance Cost.

- Disadv. :
- ① require expensive Customer Collaboration
 - ② Difficult to know how long the project will last.
 - ③ tools are expensive
 - ④ time consuming process.



Steps of this Model :

1. Requirement Gathering & Analyst
2. Quick decision
3. build a prototype
4. Assessment or user evaluation
5. Prototype refinement
6. engineer Product.

→ Evolutionary process Model : it

resemble the iterative enhancement Model. The same phases are defined for the waterfall model occurs here in a cyclical fashion.

This Model differs from the iterative enhancement Model in the sense that this does not require a useful product at end of each cycle.

In evolutionary development, requirements are implemented by category rather than by priority.

For example, in a simple database application, one cycle might implement the GUI, another file Manipulation, another queries and updates.

all 4 Cycles Must Complete before there is a working product available.

GUI allows the users to interact with the System, file Manipulation allow the data to be Saved & Retrieved, queries allow user to Get out of System and updates allows users to put data into the System.

Benefits : (1) Reduction in risk for slow projects.

(2) Reduce Costs

(3) better fit the product to user needs & Market requirements.

(4) Increase Management Visibility of project progress.